Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A particle has an initial velocity $3\hat{i}+4\hat{j}$ and an acceleration of $0.4\hat{i}+0.3\hat{j}$. Its speed after 10 sec is

(A) $7\sqrt{2}$ units

(B) 7 units

(C) 8.5 units

(D) 10 units

Q.2

The coordinates of a particle moving in $\mathrm{x}-\mathrm{y}$ plane at any instant of time t are $\mathrm{x}=4\mathrm{t}^2; \mathrm{y}=3\mathrm{t}^2$. The speed of the particle at that instant is

(A) $10\mathrm{t}$

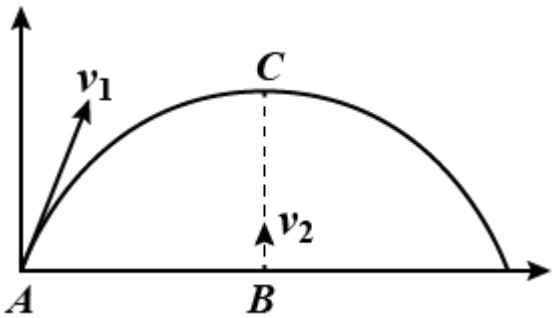
(B) $5\mathrm{t}$

(C) $3\mathrm{t}$

(D) $2t$

Q.3

A body is projected with velocity v_1 from the point A with angle of 30°, another body at the same time is projected vertically upwards from B with velocity v_2 as shown in adjoining figure. If the point B lies vertically below the highest point C, then for both bodies to collide the ratio $\frac{\mathrm{v}_2}{\mathrm{v}_1}$ should be



(A) 0.5

(B) 1

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{2}{\sqrt{3}}$

Q.4

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A projectile can have the same range R for two angles of projection. If t_1 and t_2 be the times of flight in two cases, then what is the product of two times of flight?

(A) $t_1 t_2 \propto R$

(B) $t_1 t_2 \propto R^2$

(C) $t_1 t_2 \propto 1/R$

(D) $t_1 t_2 \propto 1/R^2$

Q.5

If the angles of projection of a projectile with same initial velocity exceed or fall short of 45° by equal amounts, then the ratio of horizontal ranges is

(A) $1:2$

(B) $1:3$

(C) $1:4$

(D) $1:1$

Q.6

The equation of a projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$ The angle of projection is given by

(A) $\tan\theta = \frac{1}{\sqrt{3}}$

(B) $\tan\theta = \sqrt{3}$

(C) $\frac{\pi}{2}$

(D) zero.

Q.7

A plane flying horizontally at a height of $1500\ \mathrm{m}$ with a velocity of $200\ \mathrm{ms}^{-1}$ passes directly overhead on antiaircraft gun. Then the angle with the horizontal at which the gun should be fired from the shell with a muzzle velocity of $400\ \mathrm{ms}^{-1}$ to hit the plane, is

(A) 90°

(B) 60°

(C) 30°

(D) 45°

Q.8

The equation of trajectory of projectile is given by $\mathrm{y} = \frac{\mathrm{x}}{\sqrt{3}} - \frac{\mathrm{gx}^2}{20}$, where x and y are in metre.
The maximum range of the projectile is

(A) $\frac{8}{3}\ \mathrm{m}$

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(B) $\frac{4}{3}m$

(C) $\frac{3}{4}$ m

(D) $\frac{3}{8}$ m

Q.9

A bullet is fired with a speed of 1500 m/s in order to hit a target 100 m away. If $g = 10$ m/s^2. The gun should be aimed

(A) 15 cm above the target

(B) 10 cm above the target

(C) 2.2 cm above the target

(D) directly towards the target

Q.10

A projectile is thrown in the upward direction making an angle of 60° with the horizontal direction with a velocity of 147 ms^{-1}. Then the time after which its inclination with the horizontal is 45°, is

(A) 15 s

(B) 10.98 s

(C) 5.49 s

(D) 2.745 s

Q.11

Two pegs A and B thrown with speeds in the ratio $1:3$ acquired the same heights. If A is thrown at an angle of 30° with the horizontal, the angle of projection of B will be

(A) 0°

(B) $\sin^{-1}\left(\frac{1}{8}\right)$

(C) $\sin^{-1}\left(\frac{1}{6}\right)$

(D) $\sin^{-1}\left(\frac{1}{2}\right)$

Q.12

A body projected at an angle with the horizontal has a range 300 m. If the time of flight is 6 s, then the horizontal component of velocity is

(A) 30 m s^{-1}

(B) 50 m s^{-1}

(C) 40 m s^{-1}

(D) 45 m s^{-1}

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Q.13

A boy can throw a stone up to a maximum height of $10\ \mathrm{m}$. The maximum horizontal distance that the boy can throw the same stone up to will be

(A) $20\sqrt{2}\ \mathrm{m}$

(B) $10\ \mathrm{m}$

(C) $10\sqrt{2}\ \mathrm{m}$

(D) $20\ \mathrm{m}$

Q.14

A projectile is given an initial velocity of $(\hat{i}+2\hat{j})\mathrm{m/s}$, where $\hat{i}$ is along the ground and j is along the vertical. If $\mathrm{g}=10\ \mathrm{m/s^2}$, the equation of its trajectory is :

(A) $y=x-5x^2$

(B) $y=2x-5x^2$

(C) $4y=2x-5x^2$

(D) $4y=2x-25x^2$

Q.15

A body is projected vertically upwards with a velocity u, after time t another body is projected vertically upwards from the same point with a velocity v, where $v<u$. If they meet as soon as possible, then choose the correct option

(A) $t=\frac{u-v+\sqrt{u^2+v^2}}{g}$

(B) $t=\frac{u-v+\sqrt{u^2-v^2}}{g}$

(C) $t=\frac{u+v+\sqrt{u^2-v^2}}{g}$

(D) $t=\frac{u-v+\sqrt{u^2-v^2}}{2g}$

Q.16

A particle is projected with a velocity v such that its range on the horizontal plane is twice the greatest height attained by it. The range of the projectile is (where g is acceleration due to gravity)

(A) $\frac{4v^2}{5g}$

(B) $\frac{4g}{5v^2}$

(C) $\frac{v^2}{g}$

(D) $\frac{4v^2}{\sqrt{5}g}$

Q.17

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A gun kept on a straight horizontal road is used to hit a car travelling on the same road away from the gun at a uniform speed of $10\sqrt{2}\ \text{ms}^{-1}$. The car is at a distance of $150\ \text{m}$ from the gun when it is fired at an angle of 45° to the horizontal. With what speed should the shell be projected so that it hits the car? Take $g = 10\ \text{ms}^{-2}$.

(A) $20\ \text{ms}^{-1}$

(B) $30\ \text{ms}^{-1}$

(C) $40\ \text{ms}^{-1}$

(D) $50\ \text{ms}^{-1}$

Q.18

A ball is projected with a certain speed from a point on the ground which is at a distance of $30\ \text{m}$ from a vertical wall. If the angle of projection is 45° with the horizontal, the ball just clears the top of the wall and strikes the ground at a distance of $10\ \text{m}$ from the wall on the other side. The height of wall must be (Take $g = 10\ \text{ms}^{-2}$)

(A) $7.5\ \text{m}$

(B) $10.0\ \text{m}$

(C) $12.5\ \text{m}$

(D) $15.0\ \text{m}$

Q.19

A body is projected horizontally with a speed u from the top A of an plane ABC inclined at an angle $\theta = 30^\circ$ with the horizontal. It strikes the plane at point P. The distance AP is given by

(A) $\frac{u^2}{2g}$

(B) $\frac{3}{2}\frac{u^2}{g}$

(C) $\frac{2}{3}\frac{u^2}{g}$

(D) $\frac{4}{3}\frac{u^2}{g}$

Q.20

The equation of trajectory of a projectile is given by $y = 4x - \frac{40}{39}x^2$, where y is vertically upwards. If R is range and θ is angle of projection from horizontal, then value of $R\tan\theta$ is (x and y are in m)

(A) $3.9\ \text{m}$

(B) $15.6\ \text{m}$

(C) $11.7\ \text{m}$

(D) $14.2\ \text{m}$

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Q.1 Answer:

$7\sqrt{2}$ units

Solution:

$\vec{u} = 3\hat{i} + 4\hat{j} \quad \Rightarrow ux = 3$ **and** $uy = 4$

$\vec{a} = 0.4\hat{i} + 0.3\hat{j} \Rightarrow ax = 0.4$ **and** $ay = 0.3$

So, $V_x = ux + a_x t = 3 + 0.4 \times 10 = 3 + 4 = 7$

$\mathrm{V_y} = \mathrm{u\hat{y} + a_y t} = 4 + 0.3 \times 10 = 4 + 3 = 7$

$\vec{V} = 7\hat{i} + 7\hat{j}$

$\Rightarrow |\vec{V}| = \sqrt{7^2 + 7^2} = 7\sqrt{2}$ **units**

Q.2 Answer:

$10\mathrm{t}$

Solution:

According to the question, at any instant t, $x = 4t^2, y = 3t^2$

$\therefore v_x = \frac{dx}{dt} = \frac{d}{dt}\left(4t^2\right) = 8t$

and $v_y = \frac{dy}{dt} = \frac{d}{dt}\left(3t^2\right) = 6t$

The speed of the particle at instant t. $v = \sqrt{v_x^2 + v_y^2} = \sqrt{(8t)^2 + (6t)^2} = 10t$

Q.3 Answer:

0.5

Solution:

The two bodies will collide at the highest point if both cover the same vertical height in the same time.

So, $\frac{v_1^2 \sin^2 30^\circ}{2g} = \frac{v_2^2}{2g}$

or $\frac{v_2^2}{v_1^2} = \sin^2 30^\circ$

$\Rightarrow \frac{v_2}{v_1} = \sin 30^\circ = \frac{1}{2} = 0.5.$

Q.4 Answer:

$\mathrm{t_1 t_2 \propto R}$

Solution:

$\mathrm{t_1} = \frac{2\mathrm{u}\sin\theta}{\mathrm{g}}$ **and**

$\mathrm{t_2} = \frac{2\mathrm{u}\sin(90-\theta)}{\mathrm{g}} = \frac{2\mathrm{u}\cos\theta}{\mathrm{g}}$

$\therefore \quad \mathrm{t_1 t_2} = \frac{4\mathrm{u}^2\cos\theta\sin\theta}{\mathrm{g}^2} = \frac{2}{\mathrm{g}}\left[\frac{\mathrm{u}^2\sin 2\theta}{\mathrm{g}}\right.$

$= \frac{2}{\mathrm{g}}\mathrm{R},$

where R **is the range. Hence** $\mathrm{t_1 t_2 \propto R}$

Q.5 Answer:

$1:1$

Solution:

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For complementary angles of projection $(45^\circ + \alpha)$ and $(45^\circ - \alpha)$ with same initial velocity u, range R is same. $\theta_1 + \theta_2 = (45^\circ + \alpha) + (45^\circ - \alpha) = 90^\circ$

Q.6 Answer:

$\tan\theta = \sqrt{3}$

Solution:

Comparing the given equation with $\mathrm{y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}}$, we get
$\tan\theta = \sqrt{3}$

Q.7 Answer:

60°

Solution:

Horizontal distance covered should be same for the time of collision. $400\cos\theta = 200$ **or** $\cos\theta = \frac{1}{2}$ **or** $\theta = 60^\circ$

Q.8 Answer:

$\frac{4}{3}m$

Solution:

Comparing the given equation with the equation of trajectory of a projectile, $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$
we get, $\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$
and $2u^2\cos^2\theta = 20 \Rightarrow u^2 = \frac{20}{2\cos^2\theta}$
$= \frac{40}{3}$
Now, $\mathrm{R_{max} = \frac{u^2}{g} = \frac{40}{3\times 10} = \frac{4}{3}\ m}$

Q.9 Answer:

2.2 cm above the target

Solution:

The bullet performs a horizontal journey of 100 m **with constant velocity of** 1500 m/s**. The bullet also performs a vertical journey of h with zero initial velocity and downward acceleration** g**.**
$\therefore$ **For horizontal journey, time** $\mathrm{(t) = \frac{Distance}{Velocity}}$ $\therefore \mathrm{t = \frac{100}{1500} = \frac{1}{15}sec} \quad \ldots(1)$
The bullet performs vertical journey for this time. For vertical journey, $\mathrm{h = ut + \frac{1}{2}gt^2}$
$\mathrm{h = 0 + \frac{1}{2}\times 10\times\left(\frac{1}{15}\right)^2}$
or, $\mathrm{h = \frac{20}{9}\ cm = 2.2\ cm}$

Q.10 Answer:

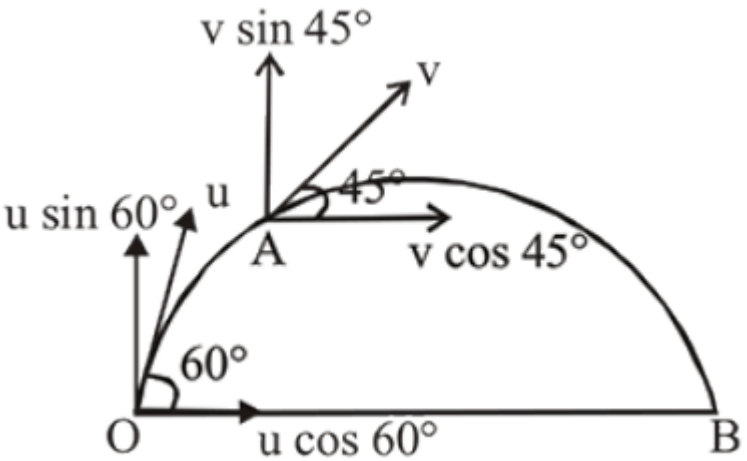
5.49 s

Solution:

Velocity of projectile $\mathrm{u = 147\ ms^{-1}}$ **angle of projection** $\alpha = 60^\circ$ **Let, the time taken by the projectile**

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from O to A be t where direction $\beta = 45^\circ$. As horizontal component of velocity remains constant during the projectile motion.



$\Rightarrow \quad v\cos 45^\circ = u\cos 60^\circ$

$\Rightarrow \quad \mathrm{v} \times \frac{1}{\sqrt{2}} = 147 \times \frac{1}{2} \Rightarrow \mathrm{v} = \frac{147}{\sqrt{2}}\ \mathrm{ms}^{-1}$

For Vertical motion, $\mathrm{v_y} = \mathrm{u_y} - \mathrm{gt}$

$\Rightarrow \quad v\sin 45^\circ = 45\sin 60^\circ - 9.8\mathrm{t}$

$\Rightarrow \frac{147}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = 147 \times \frac{\sqrt{3}}{2} - 9.8\mathrm{t}$

$\Rightarrow \quad 9.8\mathrm{t} = \frac{147}{2}(\sqrt{3}-1) \Rightarrow \mathrm{t} = 5.49\ \mathrm{s}$

Q.11 Answer:

$\sin^{-1}\left(\frac{1}{6}\right)$

Solution:

$$\text{max height, } \mathrm{H_A} = \frac{\mathrm{u_A^2}\sin^2 30^\circ}{2\,\mathrm{g}}$$

$$\mathrm{H_B} = \frac{\mathrm{u_B^2}\sin^2\theta}{2\,\mathrm{g}}$$

$$\text{As we know, } \mathrm{H_A} = \mathrm{H_B}$$

$$\frac{\mathrm{u_A^2}\sin^2 30^\circ}{2\,\mathrm{g}} = \frac{\mathrm{u_B^2}\sin^2\theta}{2\,\mathrm{g}}$$

$$\Rightarrow \frac{\sin^2\theta}{\sin^2 30^\circ} = \frac{\mathrm{u_A^2}}{\mathrm{u_B^2}}$$

$$\sin^2\theta = \left(\frac{\mathrm{u_A}}{\mathrm{u_B}}\right)^2 \sin^2 30^\circ \Rightarrow \sin^2\theta = \frac{1}{36}$$

$$\sin\theta = \frac{1}{6} \Rightarrow \theta = \sin^{-1}\left(\frac{1}{6}\right)$$

Q.12 Answer:

$50\ \mathrm{m\ s^{-1}}$

Solution:

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As we know, $\mathrm{R} = \mathrm{u}\cos\theta \times \mathrm{t}$
Given, $\mathrm{R} = 300\ \mathrm{m}, \mathrm{t} = 6\ \mathrm{s}$
$\mathrm{u}\cos\theta = \frac{\mathrm{R}}{\mathrm{t}} = \frac{300}{6} = 50\ \mathrm{ms}^{-1}$

Q.13 Answer:

$20\ \mathrm{m}$

Solution:

$R = \frac{u^2\sin^2\theta}{g}, H = \frac{u^2\sin^2\theta}{2g}$
$H_{\max}$ at $2\theta = 90^\circ$ So, $H_{\max} = \frac{u^2}{2g}$
$\frac{u^2}{2g} = 10 \Rightarrow u^2 = 10g \times 2$
$R = \frac{u^2\sin 2\theta}{g} \Rightarrow R_{\max} = \frac{u^2}{g} = 20$ metre

Q.14 Answer:

$y = 2x - 5x^2$

Solution:

From equation, $\vec{v} = \hat{i} + 2\hat{j} \Rightarrow \quad \mathrm{x} = \mathrm{t} \quad \ldots$
$y = 2t - \frac{1}{2}\left(10t^2\right)$
From (i) and (ii), $\quad y = 2x - 5x^2$

Q.15 Answer:

$t = \frac{u - v + \sqrt{u^2 - v^2}}{g}$

Solution:

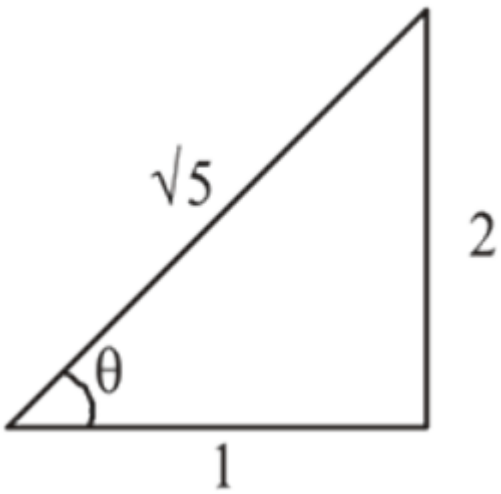
Let the two bodies meet each other at a height h after time T of the projection of second body. Then before meeting, the first body was in motion for time $(t + T)$ whereas the second body was in motion for time T. The distance moved by the first body in time $(t + T) = u(t + T) - \frac{1}{2}g(t + T)^2$
And the distance moved by the second body in time $T = vT - \frac{1}{2}gT^2 = h$ (supposed above). ..... $\because$ The two bodies meet each other, $\therefore$ They are equidistant from the point of projection.
Hence, $u(t + T) - \frac{1}{2}g(t + T)^2 = vT - \frac{1}{2}gT^2$
or $gt^2 + 2t(gT - u) + 2(v - u)T = 0 \ldots\ldots\ldots$ (2)
Also from (1) we get, $h = vT - \frac{1}{2}gT^2$
$\frac{dh}{dT} = v - gT$
h increases as T increases
T is minimum when h is minimum i.e., when
$\frac{dh}{dT} = 0$, i.e. when $v - gT = 0$ or $T = \mathrm{v/g}$. Substituting this value of T in (2), we get $gt^2 + 2t(v - u) + 2(v - u)(v/g) = 0$
or $t = \frac{2g(u-v) + \sqrt{4g^2(u-v)^2 + 8vg^2(u-v)}}{2g^2}$
or $t = \frac{u - v + \sqrt{u^2 - v^2}}{g}$
neglecting the negative sign which gives negative value of t

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Q.16 Answer:

$\frac{4v^2}{5g}$

Solution:



We know, $R = 4H\cot\theta \Rightarrow \cot\theta = \frac{1}{2}$

From triangle we can say that $\sin\theta = \frac{2}{\sqrt{5}}, \cos\theta = \frac{1}{\sqrt{5}}$

∴ Range of projectile $R = \frac{2v^2\sin\theta\cos\theta}{g}$

$= \frac{2v^2}{g} \times \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} = \frac{4v^2}{5g}$

Q.17 Answer:

$50\ \text{ms}^{-1}$

Solution:

Let at $t = 0, O$ **and** A **are the positions of the gun and the car (Fig.). Let us say that at time** $t = t_0$**, the shell and the car reach** B **simultaneously so that the shell hits the car when it is at a distance** OB **from the gun. Let** u **be the speed of projection of the shell. Then, initial horizontal component of velocity of the shell** $= u\cos 45^\circ = \frac{u}{\sqrt{2}}$**, and initial vertical component** $= u\sin 45^\circ = u/\sqrt{2}$**. Therefore, the time of flight** $= \frac{2u}{\sqrt{2}g} = \sqrt{2}u/g$**. The car takes this time to cover the distance** AB **while the shell covers the distance** OB **in this time. Now** $OB = OA + AB = 150\ \text{m} + AB$**. Distance** AB **is given by**

and $AB = 10\sqrt{2} \times \sqrt{2} \times \frac{u}{g} = 20u/g$

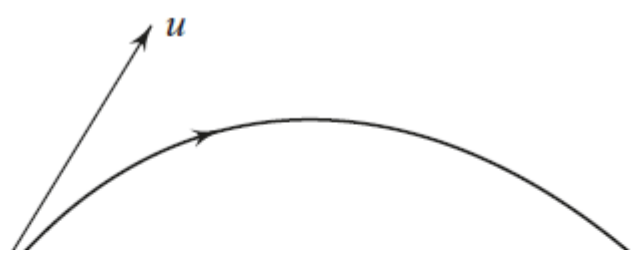
$OB = \frac{u}{\sqrt{2}} \times \frac{\sqrt{2}u}{g} = u^2/g$

$\therefore$

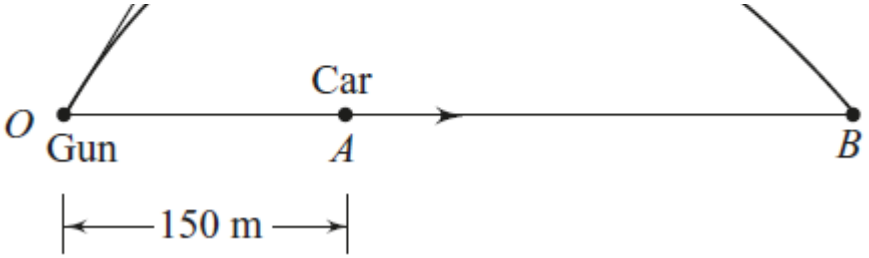
$\frac{u^2}{g} = 150 + \frac{20u}{g}$

or $\quad u^2 - 20u - 1500 = 0 \quad (\because g = 10\ \text{ms}^{-2})$

The positive root of this quadratic equation is $u = 50\ \text{ms}^{-1}$**. Hence the correct choice is** (d)**.**



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Q.18 Answer:

7.5 m

Solution:

$$R = 30 + 10 = 40 \text{ m}$$
$$\theta = 45^\circ$$
$$R = \frac{u^2}{g} = 40$$
$$u^2 = 40 \times 10$$
$$u = 20 \text{ m/s}$$
$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$
$$h = 30\tan 45^\circ - \frac{10 \times 30 \times 30}{2 \times 20 \times 20 \times \cos^2 45^\circ}$$
$$= 30 - \frac{9 \times 10^3}{8 \times 100 \times \frac{1}{2}}$$
$$= 30 - \frac{90}{4} = 30 - 225$$
$$= 7.5 \text{ m}$$

Q.19 Answer:

$\frac{4}{3}\frac{u^2}{g}$

Solution:

If the time taken by the body to strike the plane at P is t (Fig.), then the horizontal and vertical distances travelled by it are

$x = ut$ **(i)**

and $y = \frac{1}{2}gt^2$ **(ii)**

Eliminating t from (i) and and (ii),

$y = \frac{gx^2}{2u^2}$

Now $\tan\theta = \frac{AD}{DP} = \frac{y}{x}$

$\Rightarrow y = x\tan\theta$. Thus

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$$x \tan\theta = \frac{gx^2}{2u^2}$$

$$\Rightarrow \quad x = \frac{2u^2 \tan\theta}{g}$$

$$\text{Distance } AP = \sqrt{x^2 + y^2}$$

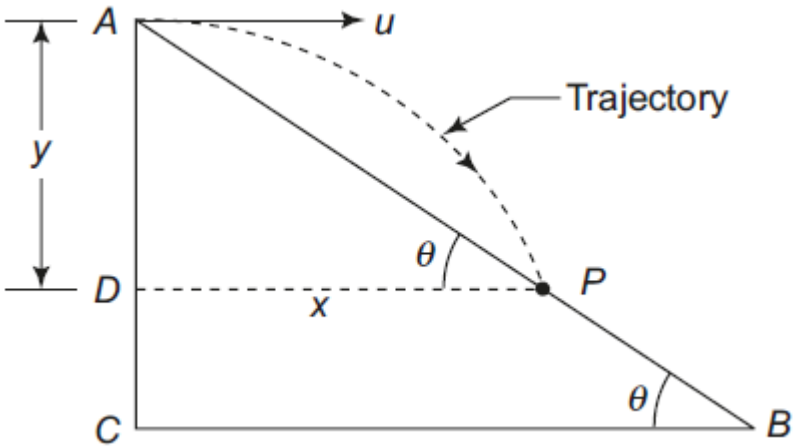
$$= \sqrt{x^2 + x^2 \tan^2\theta}$$

$$= x\sqrt{1 + \tan^2\theta}$$

$$\text{or} \quad = x \sec\theta$$

$$= \frac{2u^2 \tan\theta \sec\theta}{g}$$

$$\text{For } \theta = 30^\circ, AP = \frac{2u^2 \sin 30^\circ}{g \cos^2 30^\circ} = \frac{4u^2}{3g}$$



Q.20 Answer:

15.6 m

Solution:

$y = x \tan\theta - \frac{g^2 x^2}{2u^2 \cos^2\theta}$

Range when $y = 0$

$\tan\theta = 4$

$y = 0$

$4x = \frac{40}{39}x^2$

$x = \frac{4 \times 39}{40} = 3.9 \text{ m}$

$R = 3.9 \text{ m}$

$\tan\theta = 4$

$R \tan\theta = 4 \times 3.9 = 15.6 \text{ m}$

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